Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 1 / 方法 1 [AL02-M01]

A polynomial is a structured object: coefficients encode root symmetries, factorization controls multiplicity, and a degree bound makes identities rigid. Reduce the expression before solving roots, and restore all real-domain restrictions afterward.

多项式具有结构：系数编码根的对称性，因式分解控制重数，次数界使恒等式具有刚性。先约化表达式，再求根，最后恢复全部实数定义域限制。

Recover symmetric root expressions directly from coefficients

直接由系数恢复对称根式

Procedure

1. Normalize the leading coefficient and identify the elementary sums e_1, e_2, e_3.

2. Use $p_2 = e_1^2 - 2e_2$ and $p_3 = e_1p_2 - e_2p_1 + 3e_3$, rather than solving roots individually.

3. For two bases with sum s and product p, their power sums satisfy $u_{n+2} = su_{n+1} - pu_n$.

4. Account explicitly for repeated copies of each root, and check real-root feasibility if the original variables are real.

通用步骤

1. 先归一首项系数，再识别基本对称和 e_1, e_2, e_3。

2. 使用 $p_2 = e_1^2 - 2e_2, p_3 = e_1p_2 - e_2p_1 + 3e_3$，避免逐个解根。

3. 两个底数的和为 s、积为 p 时，其幂和满足 $u_{n+2} = su_{n+1} - pu_n$。

4. 明确处理每个根的重复副本；若原变量为实数，还须检查实根可行性。

> Example. The roots of $x^3 - 2x^2 - 3x + 1$ are r, s, t. Find $r^3 + s^3 + t^3$.

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 1 / 方法 1 [AL02-M01] (continued / 续)

> 例题：x^3-2x^2-3x+1 的根为 r,s,t，求 $r^3+s^3+t^3$。

Worked solution

1. Vieta gives $e_1=2, e_2=-3, e_3=-1$.

2. The square sum is $p_2=2^2-2(-3)=10$.

3. Thus $p_3=e_1p_2-e_2e_1+3e_3=20+6-3=23$.

例题解析

1. 韦达关系给出 $e_1=2, e_2=-3, e_3=-1$。

2. 平方和为 $p_2=2^2-2(-3)=10$。

3. 故 $p_3=e_1p_2-e_2e_1+3e_3=20+6-3=23$。

Vieta signs alternate; divide by the leading coefficient first.

韦达符号交替，须先除首项系数。

If each root z_i is repeated k times, the sum is $k\sum z_i$, while the pair sum is $\binom{k}{2}\sum z_i^2+k^2\sum_{i<j}z_iz_j$. This counts same-root copies separately from different-root copies.

For square roots s_1,s_2,s_3, the product $(s_1+s_2+s_3)(-s_1+s_2+s_3)(s_1-s_2+s_3)(s_1+s_2-s_3)$ equals $4e_2-e_1^2$, where e_i refer to s_1^2,s_2^2,s_3^2. A zero sign sum therefore imposes a symmetric coefficient relation. Still check that other sign sums do not collide.

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 1 / 方法 1 [AL02-M01] (continued / 续)

每个根 z_i 重复 k 次时，总和为 $k\sum z_i$，两两积和为 $\binom{k}{2}\sum z_i^2 + k^2\sum_{i<j} z_i z_j$；须分开同根副本与异根副本。

对平方根 s_1, s_2, s_3，乘积 $(s_1+s_2+s_3)(-s_1+s_2+s_3)(s_1-s_2+s_3)(s_1+s_2-s_3)$ 等于 $4e_2 - e_1^2$，其中 e_i 是 s_1^2, s_2^2, s_3^2 的基本对称式。零符号和因此给出系数关系，但仍须检查其他符号和是否重合。

Practice / 配套练习: AIME-2001-I-Q03 (WB p. 32), AIME-2002-I-Q06 (WB p. 117), AIME-2005-I-Q08 (WB p. 118), AIME-2008-II-Q07 (WB p. 42), AIME-2011-I-Q15 (WB p. 556), AIME-2015-II-Q06 (WB p. 48), AIME-2015-II-Q14 (WB p. 31), AIME-2019-I-Q08 (WB p. 51), AIME-2019-I-Q10 (WB p. 52), AIME-2021-II-Q07 (WB p. 28), AIME-2021-II-Q04 (WB p. 36), AIME-2026-I-Q06 (WB p. 124), AIME-2026-II-Q11 (WB p. 98), AIME-2010-II-Q07 (WB p. 71), AIME-2002-II-Q15 (WB p. 341)

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 2 / 方法 2 [AL02-M02]

Reduce a polynomial at roots and interchange root products

在根处约化多项式并交换根乘积

Procedure

1. Divide $A(x)$ by the polynomial $Q(x)$ defining the roots: at each root only the remainder matters.

2. For a sum, use Vieta or power-sum identities on the lower-degree remainder.

3. For a product, factor one polynomial and interchange the two root products, carefully tracking degree-dependent signs and leading coefficients.

通用步骤

1. 以定义根的多项式 $Q(x)$ 除 $A(x)$；在每个根处只需保留余式。

2. 求和时，对较低次余式使用韦达或幂和恒等式。

3. 求积时，分解一个多项式并交换两组根的乘积，仔细追踪次数决定的符号与首项系数。

> Example. If r, s are the roots of $x^2 - x - 1$, find $r^4 + s^4$ and $(r-2)(s-2)$.

> 例题：若 r, s 是 $x^2 - x - 1$ 的根，求 $r^4 + s^4$ 与 $(r-2)(s-2)$。

Worked solution

1. At either root $x^2 = x + 1$, so $x^4 = 3x + 2$. Hence $r^4 + s^4 = 3(r+s) + 4 = 7$.

2. The monic factorization is $Q(x) = (x - r)(x - s)$.

3. Thus $(r-2)(s-2) = Q(2) = 1$, since two sign reversals cancel.

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 2 / 方法 2 [AL02-M02] (continued / 续)

例题解析

1. 根处 $x^2 = x + 1$，故 $x^4 = 3x + 2$，从而 $r^4 + s^4 = 3(r + s) + 4 = 7$。

2. 首一因式分解为 $Q(x) = (x - r)(x - s)$。

3. 所以 $(r - 2)(s - 2) = Q(2) = 1$，两次反号相消。

At roots of Q, equality modulo Q is exact equality of values.

在 Q 的根处，模 Q 相等就是函数值精确相等。

Practice / 配套练习: AIME-2003-II-Q09 (WB p. 38), AIME-2023-II-Q08 (WB p. 84), AIME-2024-II-Q13 (WB p. 97)

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 3 / 方法 3 [AL02-M03]

Recognize hidden powers, telescoping products and convolutions

识别隐藏幂、望远镜乘积与卷积

Procedure

1. Compare coefficients with $(x-h)^k$, a geometric sum, or a product of two short coefficient lists.

2. Use difference-of-squares or difference-of-cubes identities to telescope products or rationalize a radical.

3. If a multiplier such as $x-1$ is introduced, record its zeros and remove only the roots actually introduced by it.

4. After factoring, check which factors can vanish on the requested real or complex domain.

通用步骤

1. 将系数与 $(x-h)^k$、等比和或两个短系数表的乘积比较。

2. 用平方差或立方差恒等式使乘积望远镜消去，或对根式有理化。

3. 若引入 $x-1$ 等乘子，记录其零点，并只删除由该乘子真正引入的根。

4. 分解后，检查哪些因子能在所求实数或复数定义域内为零。

> Example. Find a polynomial R with $R(0)=1$ such that $R(x)^2=1+2x+3x^2+2x^3+x^4$, and prove uniqueness.

> 例题：求满足 $R(0)=1$ 且 $R(x)^2=1+2x+3x^2+2x^3+x^4$ 的多项式 R，并证明唯一性。

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 3 / 方法 3 [AL02-M03] (continued / 续)

Worked solution

1. Degree comparison forces $R = 1 + ux + vx^2$. The coefficient of x in its square is $2u$, so $u = 1$.

2. The x^2 coefficient is $u^2 + 2v = 3$, so $v = 1$.

3. The candidate $R = 1 + x + x^2$ also gives the required x^3, x^4 coefficients. Both unknown coefficients were forced, proving uniqueness.

例题解析

1. 比较次数得 $R = 1 + ux + vx^2$；平方的 x 系数为 $2u$，故 $u = 1$。

2. x^2 系数为 $u^2 + 2v = 3$，故 $v = 1$。

3. 候选 $R = 1 + x + x^2$ 也给出所需的 x^3, x^4 系数；两个未知系数均被迫确定，故唯一。

Convolution explains a triangular coefficient pattern; inspection alone does not establish multiplicities.

卷积解释三角形系数模式；单凭观察不能确定根的重数。

Practice / 配套练习: AIME-2000-II-Q13 (WB p. 54), AIME-2003-II-Q15 (WB p. 88), AIME-2004-I-Q13 (WB p. 89), AIME-2005-I-Q06 (WB p. 68), AIME-2005-II-Q07 (WB p. 40), AIME-2006-I-Q12 (WB p. 199), AIME-2013-I-Q05 (WB p. 33), AIME-2014-I-Q09 (WB p. 46), AIME-2019-I-Q12 (WB p. 95), AIME-2024-II-Q11 (WB p. 61), AIME-2023-II-Q04 (WB p. 21), AIME-2002-II-Q11 (WB p. 154), AIME-2015-I-Q03 (WB p. 469)

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 4 / 方法 4 [AL02-M04]

Partition preimages and common roots before matching coefficients

比较系数前先划分原像与公共根

Procedure

1. A degree-d polynomial has at most d distinct roots unless it is zero.

2. For a composition or several level values, partition the inputs according to which target root or level they map to.

3. Apply Vieta within each group and compare coefficients that the defining polynomials share.

4. For shared factors, use the degree of the least common multiple, subtract equations to isolate the common root, and verify existence.

通用步骤

1. 非零 d 次多项式至多有 d 个不同根。

2. 对复合式或多个水平值，按输入映到哪个目标根或水平值来分组。

3. 在每组内使用韦达，并比较定义多项式共有的系数。

4. 对公共因子，利用最小公倍多项式的次数，相减方程隔离公共根，并验证存在性。

> Example. Two monic quadratics differ by a nonzero constant. Can they share a root? Can their two root sums differ?

> 例题：两个首一二次多项式相差非零常数，它们能有公共根吗？两组根和能不同吗？

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 4 / 方法 4 [AL02-M04] (continued / 续)

Worked solution

1. At a common root their difference would be zero, contradicting the nonzero constant. Hence no root is shared.

2. The linear coefficients are equal because only the constant changes.

3. Vieta therefore gives equal root sums, even though the two root sets are disjoint.

例题解析

1. 若有公共根，在该根处两式之差为零，与非零常数矛盾，所以不能共根。

2. 因只改变常数项，两式的一次系数相同。

3. 故韦达给出相同根和，尽管两组根集不相交。

Root multiplicity, distinctness and grouping are three separate pieces of information.

重数、互异性与分组是三条不同信息。

Practice / 配套练习: AIME-2007-I-Q08 (WB p. 41), AIME-2014-II-Q05 (WB p. 34), AIME-2015-I-Q10 (WB p. 47), AIME-2020-I-Q11 (WB p. 58), AIME-2020-I-Q14 (WB p. 59), AIME-2020-II-Q11 (WB p. 60)

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 5 / 方法 5 [AL02-M05]

Lower the degree before interpolation or coefficient elimination

插值或系数消元前先降低次数

Procedure

1. Add or subtract polynomials whose leading terms cancel; determine the actual remaining degree.

2. A degree bound plus enough values can determine a polynomial; use a root-count argument to prove uniqueness.

3. For multivariable polynomials, write the coefficient basis and solve the linear vanishing conditions before seeking universal zeros.

通用步骤

1. 将首项可消去的多项式相加或相减，并确定剩余实际次数。

2. 次数界加足够多函数值可确定多项式；以根数论证证明唯一性。

3. 多变量多项式先写系数基，并解线性消零条件，再求全体多项式共有的零点。

> Example. A polynomial P of degree at most three agrees with x^2 at $x = 0, 1, 2, 3$. Find $P(10)$.

> 例题：次数至多为三的多项式 P 在 $x = 0, 1, 2, 3$ 处与 x^2 相等，求 $P(10)$。

Worked solution

1. Form $R(x) = P(x) - x^2$, whose degree is at most three.

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 5 / 方法 5 [AL02-M05] (continued / 续)

2. The four distinct given inputs are roots of R. A nonzero polynomial of degree at most three cannot have four roots.

3. Therefore R is identically zero and $P(10) = 10^2 = 100$.

例题解析

1. 作 $R(x) = P(x) - x^2$，其次数至多为三。

2. 四个不同输入均为 R 的根，非零的至多三次多项式不可能有四个不同根。

3. 故 R 恒为零，$P(10) = 10^2 = 100$。

A canceled leading term changes the degree used in Vieta as well as interpolation.

首项消去既改变插值次数，也改变韦达所使用的次数。

Practice / 配套练习: AIME-2001-I-Q03 (WB p. 32), AIME-2003-II-Q08 (WB p. 143), AIME-2008-I-Q13 (WB p. 55), AIME-2022-I-Q01 (WB p. 37)

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 6 / 方法 6 [AL02-M06]

Use integer roots and coefficient bounds as a finite algebraic count

用整数根与系数界作有限代数计数

Procedure

1. For a monic integer polynomial, rational roots are integers. Factorization into monic rational factors preserves integrality of their coefficients by Gauss's lemma.

2. Convert root sums and products into integer factor pairs or bounded root parameters.

3. When counting exactly one distinct integer solution, distinguish a repeated root from a second root coinciding with an already excluded input.

4. Divide by root permutations only after treating repeated roots; include any unrestricted coefficients as independent factors.

通用步骤

1. 首一整系数多项式的有理根为整数；由高斯引理，其首一有理因子也具有整系数。

2. 把根和与根积转成整数因子对或有界根参数。

3. 恰数一个不同整数解时，区分重根与第二个根恰等于已排除输入。

4. 处理重根后才能除以根的排列数；无约束系数作为独立因子计入。

> Example. Count monic quadratics with nonnegative integer roots and constant term 12.

> 例题：统计根均为非负整数、常数项为 12 的首一二次多项式。

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 6 / 方法 6 [AL02-M06] (continued / 续)

Worked solution

1. Write $(x-r)(x-s)$ with $rs=12$ and $0 \le r \le s$.

2. The unordered positive factor pairs are $(1,12),(2,6),(3,4)$. Zero cannot occur because the product is positive.

3. They give three distinct sums $13, 8, 7$, hence three distinct polynomials.

例题解析

1. 写为 $(x-r)(x-s)$，其中 $rs=12, 0 \le r \le s$。

2. 无序正因子对为 $(1,12),(2,6),(3,4)$；积为正，所以零不能出现。

3. 三对给出不同根和 $13, 8, 7$，所以有三个不同多项式。

Counting root lists is not automatically the same as counting polynomials.

统计根列表并不自动等于统计多项式。

Practice / 配套练习: AIME-2005-II-Q13 (WB p. 553), AIME-2010-II-Q06 (WB p. 44), AIME-2010-II-Q10 (WB p. 713), AIME-2013-I-Q10 (WB p. 76), AIME-2013-II-Q12 (WB p. 56), AIME-2014-I-Q06 (WB p. 45), AIME-2018-I-Q01 (WB p. 35), AIME-2023-I-Q09 (WB p. 53)

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 7 / 方法 7 [AL02-M07]

Use nonnegativity, discriminants and repeated roots at extrema

用非负性、判别式与最值重根

Procedure

1. A real polynomial nonnegative on an open interval has even multiplicity at each interior zero.

2. For a quadratic, use discriminant and sign conditions; for two equal global minima, factor the shifted polynomial by both double roots.

3. Compare coefficients to derive parameter conditions, then prove the roots are distinct, inside the domain and attain the claimed bound.

4. For constrained positive roots, establish compactness first and inspect repeated-root boundary configurations.

通用步骤

1. 实多项式在开区间非负，则每个内部零点的重数为偶数。

2. 二次式用判别式与符号条件；若有两个相等全局最小值，则将平移后的多项式按两个二重根分解。

3. 比较系数得到参数条件，再证明根互异、位于定义域内并达到所求界。

4. 受约束正根的最值，先建立紧致性，再检查重根边界构型。

> Example. A real quadratic Q satisfies $Q(x) \geq x^2$ for every real x and $Q(2) = 4$. Show the form of Q.

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 7 / 方法 7 [AL02-M07] (continued / 续)

> 例题：实二次式 Q 对每个实数 x 满足 $Q(x) \geq x^2$，且 $Q(2) = 4$，求 Q 的形式。

Worked solution

1. The polynomial $D(x) = Q(x) - x^2$ is nonnegative and vanishes at two.

2. A nonzero linear term at that zero would change sign across it. Thus the zero has even multiplicity, or D is identically zero.

3. Therefore $Q(x) = x^2 + c(x-2)^2$ with $c \geq 0$, including $c = 0$. Every such polynomial satisfies the requirements.

例题解析

1. 多项式 $D(x) = Q(x) - x^2$ 非负，且在二处为零。

2. 若零点只有非零一次项，越过它就会变号，所以零点重数须为偶数，或 D 恒为零。

3. 因此 $Q(x) = x^2 + c(x-2)^2$，其中 $c \geq 0$，包括 $c = 0$；每个这种多项式都满足要求。

Repeated-root conditions are necessary at an interior extremum; verify sufficiency and endpoint alternatives.

内部最值的重根条件是必要条件，仍须检验充分性与端点情形。

For positive a, b, c with fixed pair sum and product, an extremum of $a+b+c$ occurs with two variables equal: if the three were distinct positive roots of $t^3 - st^2 + qt - r$, small changes of s would retain three distinct positive roots by continuity, contradicting an extremum. This argument applies only after compactness and positivity exclude escape to the boundary.

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 7 / 方法 7 [AL02-M07] (continued / 续)

To prove four real roots of a quartic, find four disjoint sign-change intervals, counting any boundary root by its nonzero derivative. To prove there are fewer, combine a positive region with the number of derivative zeros on the remaining region.

正数 a, b, c 的两两积和与积固定时，$a+b+c$ 的最值至少有两个变量相等：若三者为 $t^3 - st^2 + qt - r$ 的三个不同正根，连续性使 s 小幅变化后仍保留三个不同正根，与最值矛盾。但此前必须以紧致性及正性排除逃向边界。

证明四次式有四个实根，可找四个不相交变号区间；若根落在区间边界，可用非零导数判断变号。证明不足四根，则可把恒正区域与其余区域的导数零点数结合。

Practice / 配套练习: AIME-2005-II-Q15 (WB p. 419), AIME-2010-I-Q06 (WB p. 43), AIME-2014-I-Q14 (WB p. 30), AIME-2018-II-Q06 (WB p. 50), AIME-2021-I-Q15 (WB p. 420), AIME-2024-I-Q15 (WB p. 189), AIME-2025-I-Q11 (WB p. 180), AIME-2025-II-Q15 (WB p. 191), AIME-2026-II-Q08 (WB p. 318)

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 8 / 方法 8 [AL02-M08]

Exploit polynomial identities through forced roots and finite root orbits

用被迫根与有限根轨道处理多项式恒等式

Procedure

1. Compare degrees and leading coefficients before substituting special inputs.

2. Use zeros of multipliers to force specific polynomial roots, then factor them out as a polynomial identity.

3. If a root is mapped to another root, analyze its finite orbit; a modulus that grows without bound contradicts a finite root set.

4. A polynomial with a nonzero real period is constant, since its finite difference otherwise has degree one less with nonzero leading coefficient.

通用步骤

1. 代入特殊值之前，先比较次数与首项系数。

2. 利用乘子零点强迫特定根，再按多项式恒等式将其提出。

3. 若根被映为另一个根，分析其有限轨道；模长无限增长将与有限根集矛盾。

4. 具有非零实周期的多项式必为常数，否则有限差分将具有低一次的次数与非零首项系数。

> Example. If a nonzero polynomial has every root on or inside the unit circle and constant term equal in modulus to its leading coefficient, prove every root has modulus one.

> 例题：若非零多项式的全部根位于单位圆内或圆上，且常数项与首项系数模长相等，证明所有根的模长均为一。

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 8 / 方法 8 [AL02-M08] (continued / 续)

Worked solution

1. By Vieta, the product of all root moduli is the modulus of the constant-to-leading coefficient ratio, namely one.

2. Every factor is at most one. If any were strictly below one, the product would be strictly below one.

3. Hence all moduli equal one. In particular no root is zero.

例题解析

1. 由韦达，所有根模长之积为常数项与首项系数比的模长，即一。

2. 每个因子均不超过一；若某个严格小于一，乘积就会严格小于一。

3. 故所有模长均为一，特别地不存在零根。

A root-map argument must show the map remains in the same finite root set.

根映射论证须证明映射始终留在同一有限根集中。

Practice / 配套练习: AIME-2007-II-Q14 (WB p. 211), AIME-2016-I-Q11 (WB p. 57)

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 9 / 方法 9 [AL02-M09]

Extract coefficients through evaluations and symmetric selections

通过代值与对称选取提取系数

Procedure

1. The coefficient of x^k in a product is the sum of all selections whose degrees total k.

2. For pair selections use $\sum_{i<j} u_i u_j = [(\sum u_i)^2 - \sum u_i^2]/2$.

3. An evaluation at one sums coefficients. To sum absolute coefficients, first prove a substitution makes all coefficients nonnegative.

通用步骤

1. 乘积中 x^k 系数等于全部所选项次数合计为 k 的选取之和。

2. 两两选取可用 $\sum_{i<j} u_i u_j = [(\sum u_i)^2 - \sum u_i^2]/2$。

3. 代入一可求系数和；若求绝对值和，须先证明某个代换使所有系数非负。

> Example. Find the sum of the absolute coefficients of $(1-x)^3(1+x^2)$.

> 例题：求 $(1-x)^3(1+x^2)$ 的系数绝对值之和。

Worked solution

1. Replacing x by $-x$ gives $(1+x)^3(1+x^2)$, whose coefficients are all nonnegative.

2. The replacement multiplies the degree-j coefficient by $(-1)^j$, so these nonnegative coefficients are the original absolute coefficients.

3. Evaluate at one: the required sum is $2^3 \cdot 2 = 16$.

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 9 / 方法 9 [AL02-M09] (continued / 续)

例题解析

1. 以 $-x$ 替换 x 得 $(1+x)^3(1+x^2)$，各系数均非负。

2. 该替换把 j 次系数乘 $(-1)^j$，故这些非负系数恰为原系数绝对值。

3. 代入一，所求和为 $2^3 \cdot 2 = 16$。

Evaluating at minus one gives an absolute sum only after a coefficient-sign proof.

只有证明系数符号规律后，代入负一才给出绝对值和。

Practice / 配套练习: AIME-2004-I-Q07 (WB p. 39), AIME-2016-II-Q06 (WB p. 49)